

Spintronic Methods

September 18, 2026

Contents

1	Notation and Conventions	2
2	Representative Material Parameter Sets	3
2.1	Property of Heavy Metal	3
2.2	Ferromagnet: CoFeB	3
2.3	Ferrimagnet: GdFeCo	3
2.4	Noncollinear Antiferromagnet: Mn ₃ Sn	4
2.5	Layered van der Waals Antiferromagnets	5
2.5.1	CrPS ₄ micromagnetic set	5
2.5.2	Bilayer CrI ₃ staggered-SOT set	7
2.5.3	(Fe, Co) ₃ GaTe ₂	7
3	Spin Hamiltonians	8
3.1	Atomistic Spin Hamiltonian	8
3.2	Macrospin Effective Fields	9
3.2.1	One-Sublattice Model	9
3.2.2	Two-Sublattice Collinear Model	11
4	Magnetization Dynamics	11
5	Thermal Stability	12
5.1	Thermal Stability Factor	12
6	Spintronics Concepts	13
6.1	Antiferromagnets	13
6.1.1	A-Type Antiferromagnet	13
7	Symmetry for Spintronics	14
7.1	Spatial Transformations and Rotation Matrices	15
7.2	Polar and Axial Vectors	17
7.3	Time-Reversal Symmetry	17
7.4	Parity or Spatial-Inversion Symmetry	18
7.5	Mirror Symmetry	18
7.6	Spin-Rotation Symmetry	18
7.7	Combined Parity–Time Symmetry in a Two-Sublattice Magnet	19
7.8	Mini-Tutorial: From $\mathcal{P}$, $\mathcal{T}$, and $\mathcal{PT}$ to Spintronic Effects	19
7.8.1	Rashba Spin–Orbit Coupling: Break $\mathcal{P}$, Keep $\mathcal{T}$	20
7.8.2	Dzyaloshinskii–Moriya Interaction: Break Bond-Center Inversion	20
7.8.3	Spin Hall and Anomalous Hall Effects: Similar Geometry, Different $\mathcal{T}$ Requirement	20

7.8.4	Edelstein Effect: Uniform Versus Staggered Spin Accumulation	21
7.8.5	Spin–Orbit Torque: Convert a Symmetry-Allowed Spin Response into Torque	21
7.8.6	Spin-Transfer Torque: No Intrinsic Inversion-Breaking Requirement . . .	22
7.8.7	Altermagnetic Spin Splitting: Break $\mathcal{T}$ Without $\mathcal{PT}$ Degeneracy	22
7.8.8	Selection-Rule Summary	23
7.9	Practical Symmetry Test	23

1 Notation and Conventions

Unless a section explicitly states otherwise, vectors are bold, unit vectors carry a hat, time is measured in seconds, and magnetic fields $\mathbf{H}$ are expressed in A m^{-1} . The following symbols are used throughout the library.

$\mathbf{m}$ Normalized magnetization, with $\|\mathbf{m}\| = 1$.

$\mathbf{s}_i$ Unit spin direction at site i .

M_s Saturation magnetization of the ferromagnetic layer.

$\mathbf{H}_{\text{eff}}$ Effective magnetic field.

$\boldsymbol{\sigma}$ Spin-polarization unit vector.

α Dimensionless Gilbert damping parameter.

γ Positive gyromagnetic ratio, $\gamma = g\mu_B/\hbar$, where g is the Landé g -factor.

γ_0 Positive gyromagnetic factor for an $\mathbf{H}$ -field formulation, $\gamma_0 = \mu_0 |\gamma|$.

θ_{SH} Spin-Hall angle.

J_c Charge-current density.

t_{FM} Ferromagnetic-layer thickness.

β Signed torque ratio, $\beta = \tau_{\text{FLT}}/\tau_{\text{DLT}}$.

$\tau_{\text{DLT}}, \tau_{\text{FLT}}$ Damping-like and field-like torque amplitudes, respectively; both have field units.

e Positive elementary charge.

μ_0 Vacuum permeability.

μ_B Bohr magneton.

$\hbar$ Reduced Planck constant.

k_B Boltzmann constant.

T Absolute temperature.

Physical constants. The following values reproduce `constantfile.m`:

$$\begin{aligned}
\gamma &= 1.760859644 \times 10^{11} \text{ rad s}^{-1} \text{ T}^{-1}, & e &= 1.602176565 \times 10^{-19} \text{ C}, \\
\mu_0 &= 1.25663706143592 \times 10^{-6} \text{ T m A}^{-1}, & \mu_B &= 9.274 \times 10^{-24} \text{ J T}^{-1}, \\
\hbar &= 6.58211951440 \times 10^{-16} \text{ eV s}, & k_B &= 1.38064852 \times 10^{-23} \text{ J K}^{-1}.
\end{aligned}$$

Convention warning. Some references define the effective field as $\mathbf{B}_{\text{eff}}$ in tesla and use $|\gamma|$, whereas this library uses $\mathbf{H}_{\text{eff}}$ in A m^{-1} and $\gamma_0 = \mu_0 |\gamma|$. Check this convention before copying an equation into simulation code.

2 Representative Material Parameter Sets

The values below reproduce particular numerical models from cited papers and repository implementations. They are model-specific parameter sets rather than universal material constants. In particular, quantities associated with a heavy-metal layer, spin polarization, device geometry, or numerical integration are identified separately from intrinsic magnetic parameters. Entries that do not yet have a source within the approved project reference set are retained provisionally and carry a visible reference warning.

2.1 Property of Heavy Metal

The spin-Hall angle belongs to the heavy-metal spin-current source and is not specific to a ferromagnetic, ferrimagnetic, or antiferromagnetic target layer. For the signed representative values in Table 1, we use $\mathbf{J}_s = \theta_{\text{SH}} \boldsymbol{\sigma} \times \mathbf{J}_c$ and choose Pt as the positive reference; Ta and W then have negative signs [5, 6, 7]. Reversing the current, layer-normal, or spin-polarization convention requires a corresponding sign change in the torque implementation.

Property	Value	Evidence and qualification
Pt spin-Hall angle	+0.07	Representative literature value quoted by Liu et al. and Pai et al.; defines the positive reference sign
High-resistivity β -Ta spin-Hall angle	−0.12 to −0.15	Consistent magnitudes from ST-FMR, perpendicular switching, and three-terminal switching
High-resistivity β -W spin-Hall angle	−0.30±0.02 (ST-FMR); −0.33±0.06 (switching)	Negative sign verified relative to Pt; the ST-FMR value is a lower bound if spin diffusion suppresses the measured torque
Heavy-metal resistivity	$\rho_{\text{HM}} = 200 \times 10^{-8} \text{ } \Omega \text{ m}$	Representative Ta/W value from the compact model; $\rho_{\text{HM}} = 200 \text{ } \mu\Omega \text{ cm}$

Table 1: Representative properties of Pt, β -Ta, and β -W heavy metals [5, 6, 7].

The compact-model README compares reported resistivities of $190 \text{ } \mu\Omega \text{ cm}$ for Ta and $170\text{--}260 \text{ } \mu\Omega \text{ cm}$ for W, then uses the representative value $\rho_{\text{HM}} = 200 \text{ } \mu\Omega \text{ cm}$ [5].

2.2 Ferromagnet: CoFeB

The CoFeB free layer is represented by a one-sublattice macrospin LLGS model with perpendicular anisotropy. The anisotropy field and saturation magnetization below are the defaults of the SOT/STT compact model; the damping constant comes from the model of Zhang et al. [5, 4].

Parameter justification. Based on Nat. Mater. **9**, 721 (2010)., $H_{\text{an}} = 2(K_{\text{bulk}} + K_i/t_{\text{FL}})/M_s$ with $K_{\text{bulk}} = 2.245 \times 10^5 \text{ J m}^{-3}$, $K_i = 1.286 \times 10^{-3} \text{ J m}^{-2}$, $M_s = 1.58 \text{ T} = 1257 \text{ emu cm}^{-3}$, when we use $t_{\text{FL}} = 1.2 \text{ nm}$, it gives $H_{\text{an}} = 2.06 \text{ T}$. Here we approximate $M_s = 1000 \text{ emu cm}^{-3}$, $H_k = 1.5 \text{ T}$ [5].

2.3 Ferrimagnet: GdFeCo

Zhang et al. use a random two-dimensional atomistic $(\text{FeCo})_{1-x}\text{Gd}_x$ lattice driven by damping-like STT, whereas Zhu et al. use a one-dimensional atomistic GdFeCo model driven by SOT [14,

Quantity	Value	Role or convention
Uniaxial anisotropy field	$B_k = 1.5 \text{ T}$	Compact-model parameter $\mathbf{H}_k$; $H_k \approx 1.19 \times 10^6 \text{ A m}^{-1}$
Saturation magnetization	$M_s = 1000 \text{ emu cm}^{-3}$	$M_s = 1.00 \times 10^6 \text{ A m}^{-1}$; $\mu_0 M_s \approx 1.26 \text{ T}$
Gilbert damping	$\alpha = 0.02$	Magnetic model parameter

Table 2: Selected CoFeB macrospin parameters from the SOT/STT compact model and Zhang et al. [5, 4].

15]. Both are fixed-length LLG or LLGS models. The 2022 paper states that its parameters are otherwise the same as those in the 2020 paper, but replaces the single exchange coefficient with three bond-specific values. Table 3 therefore keeps the two published exchange parameterizations separately rather than selecting or averaging them.

Quantity	Zhu et al. (2020)	Zhang et al. (2022)
Spatial model	1D atomistic lattice with lattice constant $d = 0.4 \text{ nm}$; strip length and width are 400 and 40 nm	Random 2D square lattice; each interior site has four nearest neighbors; the main phase diagrams use 100 atoms
Exchange	$A_{\text{ex}} = +3 \text{ meV} = +4.8065 \times 10^{-22} \text{ J}$	$A_{\text{Gd-Gd}} = -1.26 \times 10^{-21} \text{ J}$; $A_{\text{FeCo-FeCo}} = -2.83 \times 10^{-21} \text{ J}$; $A_{\text{FeCo-Gd}} = +1.09 \times 10^{-21} \text{ J}$
Perpendicular anisotropy	$K_{\text{TM}} = K_{\text{RE}} = 0.04 \text{ meV}$	same as Zhu et al. (2020)
Hard-axis anisotropy	$\kappa_{\text{TM}} = \kappa_{\text{RE}} = 0.2 \mu\text{eV}$	Not included in the stated Hamiltonian
DMI	$D = 0.128 \text{ meV}$	Not included in the stated Hamiltonian
Gilbert damping	$\alpha_{\text{TM}} = \alpha_{\text{RE}} = 0.015$	same as Zhu et al. (2020)
Landé factors	$g_{\text{TM}} = 2.2$; $g_{\text{RE}} = 2.0$	same as Zhu et al. (2020)
Sublattice magnetizations	$M_{s,\text{TM}} = 1149 \text{ kA m}^{-1}$; $M_{s,\text{RE}} = 1012 \text{ kA m}^{-1}$ at $T = 300 \text{ K}$	same as Zhu et al. (2020)
Ferrimagnet thickness	$t_{\text{FiM}} = 0.4 \text{ nm}$	same as Zhu et al. (2020)
Drive parameter	Spin-Hall angle $\theta_{\text{SH}} = 0.2$; field-like to damping-like ratio β is varied rather than fixed in the baseline list	STT efficiency η is introduced but no numerical value is stated
Compensation composition	Not stated as a composition in the parameter block	$x_{\text{MC}} = 0.23$; $x_{\text{AMC}} = 0.21$

Table 3: Atomistic GdFeCo parameter sets retained in their native model conventions [15, 14].

The exchange values are deliberately not converted into a single common coefficient. The 2020 value belongs to a one-dimensional nearest-neighbor Hamiltonian, while the 2022 values distinguish the chemical identity of each bond in a random two-dimensional alloy. Their lattice topology and bond statistics must be retained with the numerical values when either set is used.

2.4 Noncollinear Antiferromagnet: Mn_3Sn

The three-sublattice macrospin switching model of Xu et al. and the later spatially resolved atomistic domain-wall model use the same magnetic moment, DMI, anisotropy, damping, and

spin-Hall angle, but different exchange coefficients [8, 10]. The values must therefore remain attached to their model and interaction convention. Both Hamiltonians take positive A to favor antiparallel alignment, and the crystalline easy axes point toward the nearest Sn atoms. The atomistic domain-wall paper additionally states that its positive D favors clockwise order.

Quantity		Value	Model scope or convention
Mn moment		$m_s = 3\mu_B$	Per Mn site in both models
Nearest-neighbor exchange	ex-	$A = 17.53$ meV	Three-sublattice macrospin switching model
Nearest-neighbor exchange	ex-	$A = 9$ meV	Spatial atomistic domain-wall model
DMI coefficient		$D = 0.833$ meV	Same-layer interaction in both models
Crystalline anisotropy		$K = 0.196$ meV	Single-site Hamiltonian coefficient in both models
Gilbert damping		$\alpha = 0.003$	Same in both fixed-length models
Unit-cell volume		$V_{\text{cell}} = 0.3778$ nm ³	Macrospin model; $M_s = 6m_s/V_{\text{cell}} \approx 4.42 \times 10^5$ A m ⁻¹
Mn ₃ Sn thickness		$d = 30$ nm	Macrospin SOT model geometry
Time step		$\Delta t = 5$ fs	Fourth-order Runge–Kutta macrospin integration
Nanoribbon length		5.7 μm	Spatial atomistic domain-wall model geometry

Table 4: Mn₃Sn parameter sets used in the macrospin switching and atomistic domain-wall models [8, 10].

The Mn₃Sn entries are not by themselves a complete crystallographic simulation specification: a reproducible atomistic calculation must also retain the source model’s lattice, site mapping, bond list, DMI-vector orientations, boundary conditions, and pair-counting convention.

2.5 Layered van der Waals Antiferromagnets

The parameter sets in this subsection describe A-type van der Waals antiferromagnets at the layer-macrospin or layer-micromagnetic level. Each ferromagnetic layer is represented by a normalized magnetization, and neighboring layers are coupled antiferromagnetically. The areal interlayer exchange $J_{\perp}$ must not be substituted directly for an atomistic bond energy or a volume exchange stiffness. Values from different source rows are kept separate because they were obtained at different temperatures or with different experimental and computational conventions.

Reference needed. The CrPS₄, MnBi₂Te₄, and CrI₃ exchange-length entries below are provisionally retained from the direct literature compilation in Wang et al. [18]; that paper is not in the project’s currently approved source set. The source-reference number printed by Wang et al. is preserved for every row so that distinct parameter determinations are not averaged or conflated.

2.5.1 CrPS₄ micromagnetic set

Reference needed. The following values reproduce the CrPS₄ micromagnetic and energy-landscape calculations of Wang et al. [18], which is not in the approved project reference set.

Material and source	Condition	d (nm)	$J_{\perp}$ (J m^{-2})	K (J m^{-3})	l_{ex} (nm); N
CrPS ₄ , ref. 30	Experiment, 22.5 K	0.61	1.58×10^{-4}	6.33×10^3	2.77; 4.5
CrPS ₄ , ref. 31	Experiment, 2 K	0.61	2.94×10^{-4}	1.74×10^4	2.27; 3.6
CrPS ₄ , ref. 16	Calculation	0.61	2.75×10^{-4}	1.26×10^4	2.58; 4.2
MnBi ₂ Te ₄ , ref. 44	Experiment, 2 K	1.36	1.90×10^{-2}	1.47×10^6	2.98; 2.2
MnBi ₂ Te ₄ , ref. 45	Experiment, 1.8 K	1.36	6.42×10^{-3}	1.45×10^6	1.73; 1.3
CrI ₃ , ref. 46	Calculation	0.66	1.96×10^{-4}	4.04×10^5	0.40; 0.6
CrI ₃ , ref. 47	Calculation	0.66	3.09×10^{-4}	4.62×10^5	0.47; 0.7

Table 5: Source-specific interlayer exchange and anisotropy sets compiled by Wang et al. [18]. Here d is the monolayer thickness, l_{ex} is the reported interlayer exchange length, and $N = l_{\text{ex}}/d$ is the number of layers within that length. Rows are matched sets and must not be mixed.

Quantity	Value	Scope or convention
Monoclinic anisotropy constants	$K_1 = 12,485 \text{ J m}^{-3}$; $K_2 = 8,897 \text{ J m}^{-3}$; $K_m = 1,271 \text{ J m}^{-3}$	Layer-resolved COMSOL micromagnetic model; corresponding $\mu_0 H_{K1} = 0.167 \text{ T}$, $\mu_0 H_{K2} = 0.119 \text{ T}$, and $\mu_0 H_{Km} = 0.017 \text{ T}$
Saturation magnetization	$M_s = 74,803 \text{ A m}^{-1}$	$\mu_0 M_s = 0.094 \text{ T}$
Interlayer exchange	$J_{\perp} = 1.58 \times 10^{-4} \text{ J m}^{-2}$	Antiferromagnetic; reported equivalent coupling field $\mu_0 H_C = 3.45 \text{ T}$
Intralayer exchange	$J_{\parallel} = 1.48 \times 10^{-2} \text{ J m}^{-2}$	Areal interaction used in the COMSOL model
Gilbert damping	$\alpha = 0.1$	Fixed-length LLG simulation input
Intralayer spin stiffness	$A = 1.5 \times 10^{-12} \text{ J m}^{-1}$	Separate domain-wall energy-landscape calculation
Uniaxial anisotropy	$K = 6.0 \times 10^3 \text{ J m}^{-3}$	Separate domain-wall energy-landscape calculation; do not substitute for the three monoclinic constants above

Table 6: CrPS₄ parameters retained in the distinct model conventions used by Wang et al. [18].

2.5.2 Bilayer CrI₃ staggered-SOT set

Reference needed. The bilayer CrI₃ values below are provisional because Xue and Haney [19] is not in the approved project reference set. This is a two-sublattice, fixed-length macrospin calculation for electron-doped antiferromagnetic bilayer CrI₃, not an experimental device parameterization.

Quantity	Value	Scope or convention
Easy-axis anisotropy field	$H_A \approx 1.77$ T	Source uses a magnetic-field-in-tesla convention
Antiferromagnetic exchange field	$H_E \approx 0.76$ T	Comparable to H_A in the source model
Gilbert damping	$\alpha \approx 0.04$	Common to both fixed-length layer macrospins
Even torkance, $\hat{N} \parallel \hat{z}$	$\sim 1 ea_0/\hbar$	At about 0.1 eV above the conduction-band minimum
Even torkance, $\hat{N} \parallel \hat{x}$	$\sim 0.4 ea_0/\hbar$	Same first-principles torque convention
Switching threshold	$ E \approx 2$ V μm^{-1}	Predicted from the interpolated angular torkance
Representative reversal	$E = -2.3$ V μm^{-1} ; < 100 ps	Damping-like torque drives reversal; field-like torque assists
Oscillatory state	$E = 3$ V μm^{-1} ; ~ 80 GHz	Requires both damping-like and field-like torque components

Table 7: Bilayer CrI₃ macrospin and first-principles SOT parameters from Xue and Haney [19].

2.5.3 (Fe, Co)₃GaTe₂

Guo et al. model each ferromagnetic FCGT layer as one fixed-length macrospin and demonstrate SOT switching in odd-layer Pt/FCGT devices. Their main article reports a normalized exchange field but not independent absolute values of H_{an} and H_{ex} ; only the published ratio is therefore listed below [17].

Quantity	Value	Scope or convention
Néel temperature	$T_N = 196$ K	Bulk FCGT characterization
Exchange-to-anisotropy field ratio	$H_{\text{ex}} = H_{\text{an}}/3$	Experimental-data-derived macrospin input; absolute fields are not stated in the main article
Applied in-plane field in model	$H_{\text{ext}} = 0.4H_{\text{an}}$	SOT-switching simulation input
Gyromagnetic ratio	$\gamma = 1.76 \times 10^{11}$ T ⁻¹ s ⁻¹	Fixed-length Gilbert-form LLG model
Gilbert damping	$\alpha = 0.1$	Selected simulation input, not a reported FCGT measurement
Critical current density	$J_c = 11.4$ MA cm ⁻²	11-layer Pt/FCGT device at 180 K
SOT efficiency	~ 38 Oe (MA cm ⁻²) ⁻¹	Estimated from current-induced hysteresis-loop shift
Spin-mixing conductance	1.93×10^{15} Ω^{-1} m ⁻²	Pt/FCGT interface inferred from spin-Hall magnetoresistance
Field-free switching ratio	$\sim 70\%$ at 120 K	Pt/FCGT/CrSBr heterostructure; partial switching is attributed to a multidomain CrSBr layer

Table 8: FCGT macrospin inputs and experimental SOT figures of merit from Guo et al. [17].

The selected sources provide no damping constant, torque coefficient, or complete LLG parameterization for MnBi₂Te₄. Table 5 therefore records only its two source-specific $J_{\perp}$ – K sets instead of filling the missing quantities from another material or model.

3 Spin Hamiltonians

3.1 Atomistic Spin Hamiltonian

Equation. Following the ASTRO exchange-sign convention, a classical spin Hamiltonian containing exchange, uniaxial anisotropy, Dzyaloshinskii–Moriya interaction, and Zeeman energy is

$$\begin{aligned} \mathcal{H} = & + \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i K_i (\mathbf{s}_i \cdot \hat{\mathbf{e}}_i)^2 \\ & - \frac{1}{2} \sum_{i \neq j} \mathbf{D}_{ij} \cdot (\mathbf{s}_i \times \mathbf{s}_j) - \sum_i \boldsymbol{\mu}_i \cdot \mathbf{B}_{\text{ext}}. \end{aligned} \quad (3.1)$$

Definitions. J_{ij} is the exchange-link energy in the ASTRO sign convention: $J_{ij} < 0$ favors parallel alignment, whereas $J_{ij} > 0$ favors antiparallel alignment. K_i is the uniaxial anisotropy energy, $\hat{\mathbf{e}}_i$ is the easy-axis direction, $\mathbf{D}_{ij}$ is the Dzyaloshinskii–Moriya vector, and $\boldsymbol{\mu}_i$ is the magnetic moment at site i .

Assumptions. Each $\mathbf{s}_i$ is a classical unit vector. The pair coefficients satisfy $J_{ji} = J_{ij}$ and $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$. The factors of 1/2 assume that the pair sums include both ordered pairs (i, j) and (j, i) . If each physical bond is instead included only once, for example in a unique-bond sum $\sum_{\langle i, j \rangle}$, both factors of 1/2 must be omitted. These are equivalent pair-counting conventions and must not be mixed.

Figure 1 collects the geometric quantities attached to one bond and distinguishes a physical bond from its two ordered appearances in the Hamiltonian sum.

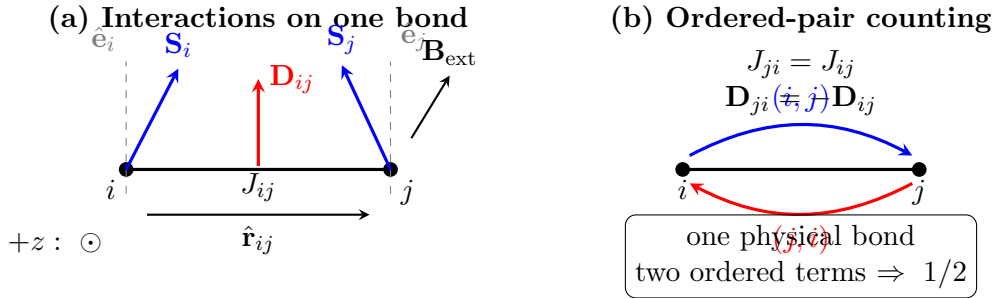


Figure 1: Original schematic of the bond quantities in Eq. (3.1). Panel (a) shows two spins, their local easy axes, the exchange link, the directed bond, the DMI vector, and the external field. The displayed DMI orientation illustrates the ASTRO convention $\mathbf{D}_{ij} = D(\hat{\mathbf{z}} \times \hat{\mathbf{r}}_{ij})$. Panel (b) shows why an ordered-pair sum contains both (i, j) and (j, i) and therefore carries a factor 1/2 [11].

Pair counting and effective field. Let $\mu_i = |\boldsymbol{\mu}_i|$ and define the effective flux density by

$$\mathbf{B}_{\text{eff},i} = -\frac{1}{\mu_i} \frac{\partial \mathcal{H}}{\partial \mathbf{s}_i}. \quad (3.2)$$

For the ordered-pair Hamiltonian above, the exchange and DMI contributions are

$$\mathbf{B}_{\text{ex},i} = -\frac{1}{\mu_i} \sum_{j \neq i} J_{ij} \mathbf{s}_j, \quad \mathbf{B}_{\text{DMI},i} = \frac{1}{\mu_i} \sum_{j \neq i} \mathbf{s}_j \times \mathbf{D}_{ij}. \quad (3.3)$$

The two ordered occurrences of each bond give identical derivatives, which cancel the Hamiltonian factors of $1/2$. Consequently, no extra factor of $1/2$ belongs in either effective-field expression.

Units. All terms in $\mathcal{H}$ have units of energy. Here the external flux density $\mathbf{B}_{\text{ext}}$ is measured in tesla, so no additional factor of μ_0 appears in the Zeeman term.

Source. For the extended classical spin Hamiltonian and Hamiltonian-derivative field convention, see Evans et al. [2].

Implementation notes. The ASTRO implementation evaluates the field, $\mathbf{B}_{\text{eff},i} = -(1/\mu_i) \partial \mathcal{H} / \partial \mathbf{s}_i$, rather than the total energy and includes both forward and backward nearest neighbors. The J_{ij} used above is therefore the coefficient stored in ASTRO's exchange-link arrays: its exchange field is $-(1/\mu_i) \sum_j J_{ij} \mathbf{s}_j$ [11]. No factor of $1/2$ should appear in the exchange or DMI field formulas because differentiation produces two identical bond contributions, cancelling the Hamiltonian factor. ASTRO's DMI component expressions correspond to $\mathbf{D}_{ij} = D(\hat{\mathbf{z}} \times \hat{\mathbf{r}}_{ij})$ and the antisymmetry $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$ used above.

3.2 Macrospin Effective Fields

Field convention. For a fixed-length macrospin with normalized magnetization $\mathbf{m}$, let w denote the magnetic energy density in J m^{-3} . The effective magnetic field and the corresponding effective flux density are [32]

$$\mathbf{H}_{\text{eff}} = -\frac{1}{\mu_0 M_s} \frac{\partial w}{\partial \mathbf{m}}, \quad \mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}. \quad (3.4)$$

Thus, an LLG equation written with $\gamma_0 = \mu_0 |\gamma|$ uses $\mathbf{H}_{\text{eff}}$ in A m^{-1} , whereas one written with $|\gamma|$ uses $\mathbf{B}_{\text{eff}}$ in tesla. The two conventions are physically equivalent but their field variables must not be mixed.

Unless stated otherwise, K_u and A_{12} are energy densities in J m^{-3} . The interfacial K_i in the combination $K_{\text{bulk}} + K_i/t_{\text{FM}}$ is instead in J m^{-2} , and every saturation magnetization is in A m^{-1} . The tensors $\mathbf{N}$ and $\mathbf{K}_{fp}$ are dimensionless.

3.2.1 One-Sublattice Model

Energy density and field decomposition. For uniaxial anisotropy with easy axis $\hat{\mathbf{u}}$, a dimensionless symmetric demagnetization tensor $\mathbf{N}$, an applied field $\mathbf{H}_{\text{ext}}$, and a dipolar field $\mathbf{H}_{\text{dip}}$ produced by another magnetic body, use the uniaxial-anisotropy convention from slides 13–14 of the EE237 Week 5 lecture notes [32] and take

$$w_1 = K_u \left[1 - (\mathbf{m} \cdot \hat{\mathbf{u}})^2 \right] + \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} - \mu_0 M_s \mathbf{m} \cdot (\mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{dip}}). \quad (3.5)$$

External field. The part of the final term in Eq. (3.5) that is proportional to $\mathbf{H}_{\text{ext}}$ is the Zeeman coupling to the applied field. Its derivative contributes $\mathbf{H}_{\text{ext}}$ directly to $\mathbf{H}_{\text{eff}}$ [32].

Uniaxial anisotropy field. Equation (3.4) gives [32]

$$\mathbf{H}_{\text{an}} = \frac{2K_u}{\mu_0 M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}, \quad \mathbf{B}_{\text{an}} = \frac{2K_u}{M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}. \quad (3.6)$$

For $\hat{\mathbf{u}} = \hat{\mathbf{z}}$, the anisotropy term reduces to $K_u(1 - m_z^2) = K_u \sin^2 \theta$, as on slide 13. Near $m_z = 1$, slide 14 therefore gives $H_{\text{an},z} = 2K_u/(\mu_0 M_s)$. For a thin film with bulk and interfacial uniaxial anisotropy, $K_u = K_{\text{bulk}} + K_i/t_{\text{FM}}$.

Figure 2 connects the magnetization geometry to the zero-field uniaxial energy barrier used in the thermal-stability factor.

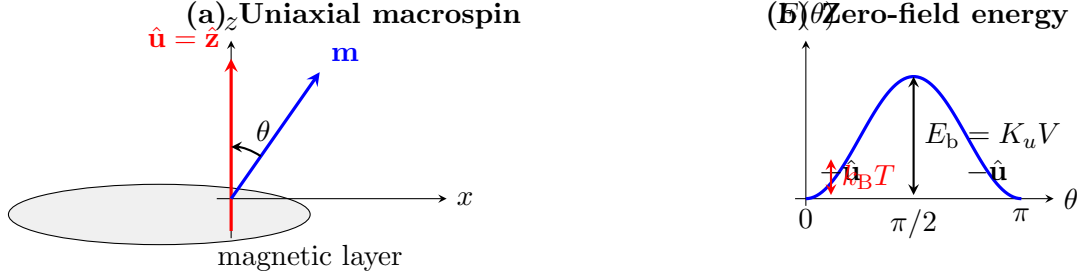


Figure 2: Original schematic linking uniaxial macrospin geometry and thermal stability. (a) The normalized magnetization $\mathbf{m}$ makes an angle θ with the easy axis $\hat{\mathbf{u}}$. (b) For a zero-field volume V with energy $E(\theta) = K_u V \sin^2 \theta$, the two easy-axis states are separated by $E_b = K_u V$; the ratio of this barrier to the thermal scale $k_B T$ is the stability factor. The conventions follow the EE237 anisotropy and thermal-stability notes [32, 33].

Demagnetization field. Slide 21 of the same notes writes the demagnetization self-energy as [32]

$$E_{\text{demag}} = -\frac{\mu_0}{2} \int_V \mathbf{M}(\mathbf{r}) \cdot \mathbf{H}_{\text{demag}}(\mathbf{r}) d^3 r, \quad (3.7)$$

$$w_{\text{demag}} = -\frac{\mu_0}{2} \mathbf{M} \cdot \mathbf{H}_{\text{demag}} = \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} \quad (\text{uniform macrospin}). \quad (3.8)$$

The factor 1/2 avoids double counting the magnet's self-interaction. Equation (3.4) gives

$$\mathbf{H}_{\text{demag}} = -M_s \mathbf{N} \mathbf{m}, \quad \mathbf{B}_{\text{demag}} = -\mu_0 M_s \mathbf{N} \mathbf{m}. \quad (3.9)$$

For principal axes, $\mathbf{N} = \text{diag}(N_x, N_y, N_z)$ and $N_x + N_y + N_z = 1$.

Dipolar field. Let a pinned macrospin have volume V_p , saturation magnetization $M_{s,p}$, and direction $\mathbf{m}_p$, so its magnetic moment is $\boldsymbol{\mu}_p = V_p M_{s,p} \mathbf{m}_p$. In the point-dipole approximation, at displacement $\mathbf{r} = r \hat{\mathbf{r}}$ from the pinned layer to the free layer, slide 22 gives the dipole formula [32]. The slide denotes it by $\mathbf{H}_i$, but its factor μ_0 makes it a flux density in tesla; therefore, in the convention of this document,

$$\mathbf{B}_{\text{dip}} = \frac{\mu_0}{4\pi r^3} [3(\boldsymbol{\mu}_p \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \boldsymbol{\mu}_p] = \mu_0 \mathbf{H}_{\text{dip}}, \quad (3.10)$$

Thermal field. Following Eq. (4) of Zhu et al. [12], write the sampled thermal field as

$$\mathbf{B}_{\text{th}} = \mu_0 \mathbf{H}_{\text{th}} = \mathcal{N}_1(0, u_B) \hat{\mathbf{x}} + \mathcal{N}_2(0, u_B) \hat{\mathbf{y}} + \mathcal{N}_3(0, u_B) \hat{\mathbf{z}}, \quad (3.11)$$

$$u_B = \sqrt{\frac{2\alpha k_B T}{V_{\text{FL}} M_s |\gamma| (1 + \alpha^2) \Delta t}}. \quad (3.12)$$

Here the $\mathcal{N}_i(0, u_B)$ are independent normal random variables with zero mean and standard deviation u_B , V_{FL} is the free-layer volume, and Δt is the duration over which a sampled field is held constant. With $|\gamma|$ in $\text{rad s}^{-1} \text{T}^{-1}$ and M_s in A m^{-1} , u_B is in tesla; the corresponding $\mathbf{H}$ -field standard deviation is $u_H = u_B/\mu_0$.

The source uses independent samples over $\Delta t = 5 \text{ ps}$ [12]. When adopting another integration interval or LLG convention, recompute the noise amplitude consistently rather than reusing that numerical value.

Total field. The total one-sublattice field, with the stochastic field added after taking the energy derivative, is [31]

$$\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{an}} + \mathbf{H}_{\text{demag}} + \mathbf{H}_{\text{dip}} + \mathbf{H}_{\text{th}}. \quad (3.13)$$

Spin-transfer and spin-orbit torques are nonconservative and are not part of the energy derivative in Eq. (3.4).

Implementation mapping. The current one-sublattice `ODE_LLГ_integral` implementation stores flux-density quantities in tesla [31]. Its `field_eta.m` evaluates the following expressions, where its input named `Hk` is B_k :

$$\begin{aligned} \mathbf{B}_{\text{an}} &= B_k m_y \hat{\mathbf{y}} \quad (\text{IMA}), & \mathbf{B}_{\text{an}} &= B_k m_z \hat{\mathbf{z}} \quad (\text{PMA}), \\ \mathbf{B}_{\text{demag}} &= -\mu_0 M_s \mathbf{N} \mathbf{m}, & \mathbf{B}_{\text{dip}} &= \mu_0 \mathbf{K}_{12} M_s \mathbf{m}_{\text{PL}}, \end{aligned}$$

after converting the input M_s from emu cm^{-3} to A m^{-1} . It sums these with the applied and thermal fields.

3.2.2 Two-Sublattice Collinear Model

Inter-sublattice exchange field. Let $\mathbf{m}_1$ and $\mathbf{m}_2$ be separately normalized sublattice magnetizations with saturation magnetizations $M_{s,1}$ and $M_{s,2}$. The exchange-energy density convention used by the two-sublattice ferrimagnetic macrospin model of Yuan et al. [13] is

$$w_{\text{ex},12} = A_{12} \mathbf{m}_1 \cdot \mathbf{m}_2, \quad A_{12} > 0, \quad (3.14)$$

Applying $\mathbf{H}_{\text{ex},i} = -(\mu_0 M_{s,i})^{-1} \partial w_{\text{ex},12} / \partial \mathbf{m}_i$ yields, for $j \neq i$,

$$\mathbf{H}_{\text{ex},i} = -\frac{A_{12}}{\mu_0 M_{s,i}} \mathbf{m}_j, \quad \mathbf{B}_{\text{ex},i} = -\frac{A_{12}}{M_{s,i}} \mathbf{m}_j. \quad (3.15)$$

Only the exchange contribution unique to the two-sublattice model is stated here; the shared external, anisotropy, demagnetization, dipolar, and thermal field formulas are defined above and are not repeated. This two-sublattice exchange extension is publication-grounded and is not implemented by the `ODE_LLГ_integral` repository.

4 Magnetization Dynamics

LLGS form with field-like torque. This macrospin equation is adapted from Eqs. (1)–(4) of Kong et al., with the atom index removed [3]. It is written in the flux-density convention: $\mathbf{B}_{\text{eff}}$ and the torque amplitudes are in tesla and are paired with γ in $\text{rad s}^{-1} \text{T}^{-1}$.

$$\begin{aligned} \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{B}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (4.1)$$

Figure 3 fixes the device axes and illustrates the vector structures that appear in Eq. (4.1).

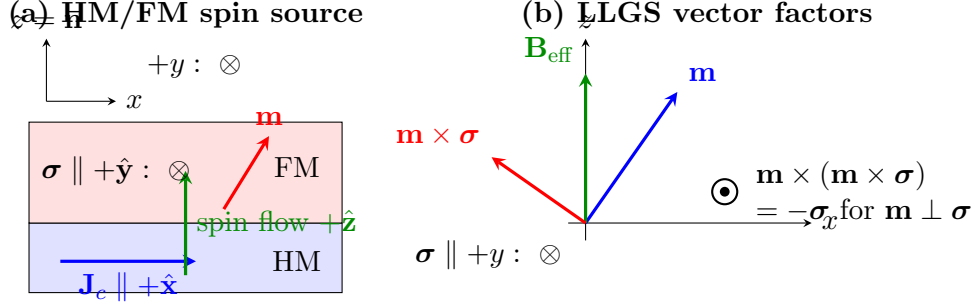


Figure 3: Original schematic of a conventional spin-Hall-driven LLGS geometry. (a) Charge current along $+x$ in the heavy metal generates a spin flow along the interface normal $+z$ with the illustrated polarization along $+y$; reversing the current or the signed spin-Hall angle reverses the polarization. (b) For $\mathbf{m}$ in the xz plane, the field-like vector $\mathbf{m} \times \boldsymbol{\sigma}$ lies in that plane, whereas the damping-like vector $\mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma})$ points along $-y$. These are vector factors only; their signed scalar prefactors are those in Eq. (4.1). The conventions follow the SOT compact model and Kong *et al.* [5, 3].

Equivalent LL form with field-like torque. The algebraically equivalent form follows the same source convention [3].

$$\begin{aligned}
 (1 + \alpha^2) \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{B}_{\text{eff}} - \alpha\gamma \mathbf{m} \times (\mathbf{m} \times \mathbf{B}_{\text{eff}}) \\
 & + \alpha\gamma\tau_{\text{DLT}} \mathbf{m} \times \boldsymbol{\sigma} - \gamma\tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\
 & + \alpha\gamma\tau_{\text{FLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) + \gamma\tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}.
 \end{aligned} \tag{4.2}$$

Torque amplitudes. For a ferromagnetic layer [4],

$$\tau_{\text{DLT}} = \frac{\hbar\theta_{\text{SH}}J_c}{2e t_{\text{FM}}M_s}, \quad \tau_{\text{FLT}} = \beta\tau_{\text{DLT}}.$$

Torque-sign convention. In the convention of Eq. (4.1), the sign product of the damping-like and field-like SOT is negative when $\beta > 0$ [1, 3]. Specifically, the scalar prefactors of their respective vector structures are $-\gamma\tau_{\text{DLT}}$ and $+\gamma\tau_{\text{FLT}}$, so their sign product is

$$(-\gamma\tau_{\text{DLT}})(+\gamma\tau_{\text{FLT}}) = -\gamma^2\beta\tau_{\text{DLT}}^2 < 0 \quad (\beta > 0).$$

Thus, a positive β means that the two SOT terms have oppositely signed scalar prefactors in the LLGS equation, even though the signed amplitudes τ_{DLT} and τ_{FLT} themselves have the same sign.

5 Thermal Stability

5.1 Thermal Stability Factor

Equation. For a single energy barrier E_b , the thermal stability factor is [33]

$$\Delta = \frac{K_u V}{k_B T} \tag{5.1}$$

The zero-field uniaxial example in Fig. 2 shows the barrier $E_b = K_u V$ relative to the thermal energy scale $k_B T$.

Units. E_b and $k_B T$ must use the same energy unit; Δ is dimensionless.

6 Spintronics Concepts

6.1 Antiferromagnets

An antiferromagnet is a magnetic material in which neighboring magnetic moments prefer an antiparallel arrangement. The opposing moments can compensate, giving little or no net magnetization even though the material has long-range magnetic order. Its order is therefore described more naturally by a Néel vector than by the net magnetization alone [9].

6.1.1 A-Type Antiferromagnet

An A-type antiferromagnet consists of ferromagnetic planes stacked antiferromagnetically: spins within each plane point in the same direction, all spins in the next plane point in the opposite direction, and the pattern repeats along the stacking direction. The word *plane* refers to the crystallographic planes selected by the exchange interactions in a particular material. Figure 4 chooses the z axis as the stacking direction as an illustrative convention, not as a universal crystal orientation.

Source status. Reference needed. This A-type structural classification was supplied directly for these notes and still needs a dedicated literature reference.

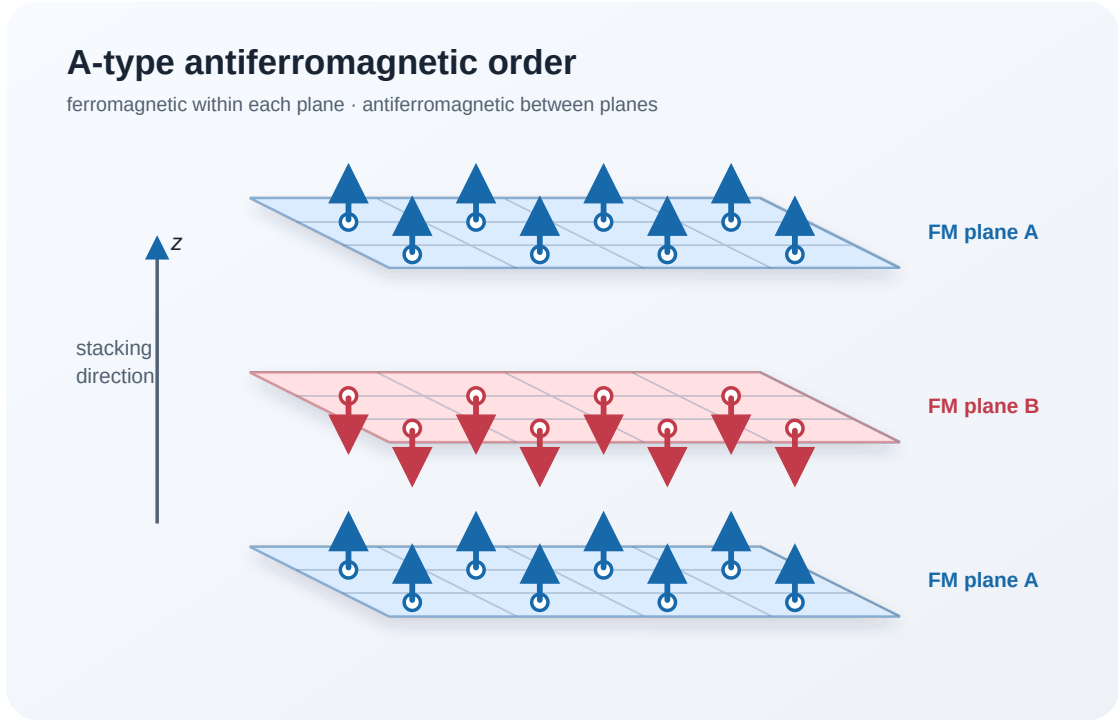


Figure 4: A-type antiferromagnetic stacking: ferromagnetic planes alternate in direction. Original schematic; the physical classification follows the provisional description above and therefore shares its **reference-needed** status.

For classical unit spins $\mathbf{s}_i$, the exchange part of the atomistic Hamiltonian can be written in the ASTRO sign convention as [11]

$$\mathcal{H}_{\text{ex}} = \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j. \quad (6.1)$$

In this convention, A-type order is favored by

$$J_{\parallel} < 0, \quad J_{\perp} > 0, \quad (6.2)$$

where $J_{\parallel}$ couples neighbors within a plane and $J_{\perp}$ couples neighbors in adjacent planes. Thus the in-plane bonds favor parallel alignment, whereas the interplane bonds favor antiparallel alignment. Some publications place a minus sign in front of the exchange sum; their signs of J are reversed. The sign convention and pair counting in Eq. (6.1) are the same as those used in the full atomistic Hamiltonian in Eq. (3.1) and in ASTRO [11].

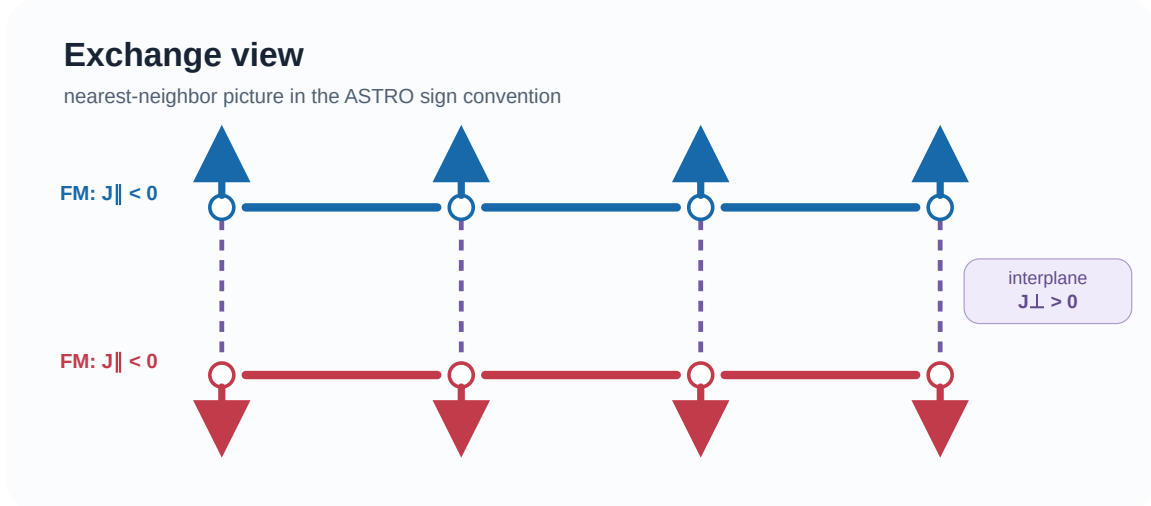


Figure 5: Exchange pattern for A-type order: parallel bonds within a layer and antiparallel bonds between layers. Original schematic based on the cited ASTRO exchange-sign convention [11]; the A-type classification itself remains **reference needed**.

For two equivalent alternating layers with normalized layer magnetizations $\mathbf{m}_A$ and $\mathbf{m}_B$, define the net and Néel order parameters as [9]

$$\mathbf{m} = \frac{\mathbf{m}_A + \mathbf{m}_B}{2}, \quad \mathbf{n} = \frac{\mathbf{m}_A - \mathbf{m}_B}{2}. \quad (6.3)$$

For an ideal compensated A-type state, these definitions give

$$\mathbf{m}_B = -\mathbf{m}_A, \quad \mathbf{m} = \mathbf{0}, \quad |\mathbf{n}| = 1. \quad (6.4)$$

The defining feature is therefore not simply zero magnetization. It is the specific spatial pattern of ferromagnetic order in two directions and antiferromagnetic alternation in the third. This distinguishes A-type order from G-type antiferromagnetism, in which every nearest neighbor is antiparallel, and C-type order, in which ferromagnetic chains are coupled antiferromagnetically. **Reference needed** for this final classification comparison.

7 Symmetry for Spintronics

A symmetry operation transforms the spatial coordinates, momenta, spins, and external fields of a system. It is a symmetry of a Hamiltonian only when the transformed Hamiltonian describes the same physical system. For a unitary operation G this condition is

$$G\mathcal{H}G^{-1} = \mathcal{H}, \quad (7.1)$$

whereas time reversal is antiunitary and also complex-conjugates scalar coefficients.

Source status. Reference needed. The general time-reversal and spatial-inversion operator rules in this section are included provisionally because no approved general reference is currently available. The mirror rules are supported by the EE237 notes, the isotropic-exchange example by ASTRO, and the magnetic parity–time example by Zhang *et al.*, as cited locally below.

7.1 Spatial Transformations and Rotation Matrices

This section uses column vectors and active transformations: the vector moves while the coordinate axes remain fixed. By the right-hand rule, a positive rotation about z is counterclockwise when viewed from the positive z axis. After a rotation through θ , the Cartesian basis vectors become

$$\begin{aligned} R_z(\theta)\hat{\mathbf{x}} &= \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix}, \\ R_z(\theta)\hat{\mathbf{y}} &= \begin{pmatrix} \cos(\frac{\pi}{2} + \theta) \\ \sin(\frac{\pi}{2} + \theta) \\ 0 \end{pmatrix} = \begin{pmatrix} -\sin \theta \\ \cos \theta \\ 0 \end{pmatrix}, \\ R_z(\theta)\hat{\mathbf{z}} &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned} \tag{7.2}$$

To see why, note that a unit vector in the xy plane at azimuth ϕ has Cartesian components $(\cos \phi, \sin \phi, 0)^\top$. Initially, $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ have azimuths 0 and $\pi/2$, respectively. An active rotation through θ adds θ to each azimuth, giving θ and $\pi/2 + \theta$; the identities $\cos(\pi/2 + \theta) = -\sin \theta$ and $\sin(\pi/2 + \theta) = \cos \theta$ then give Eq. (7.2). The vector $\hat{\mathbf{z}}$ is unchanged because it lies along the rotation axis [34]. Figure 6 shows the corresponding Cartesian projections.

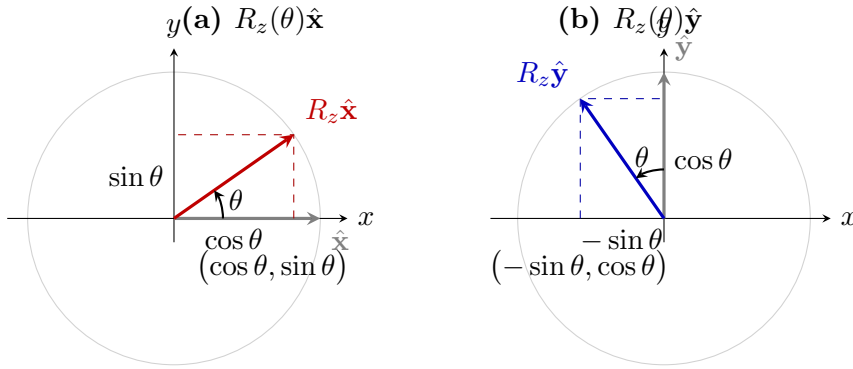


Figure 6: Original schematic showing how the Cartesian components of the actively rotated in-plane basis vectors are their projections onto the fixed axes. Both vectors rotate through the same angle θ and remain perpendicular; $\hat{\mathbf{z}}$, directed out of the page, is unchanged. The active-rotation convention follows the EE237 symmetry notes [34].

The columns of a transformation matrix are the transformed basis vectors. Therefore,

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_z(\theta) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \\ z \end{pmatrix}. \tag{7.3}$$

These conventions and the spatial transformation rules below follow the EE237 symmetry notes [34].

A passive rotation instead rotates the coordinate axes while leaving the physical vector fixed. It is represented by the inverse active rotation,

$$R_z^{-1}(\theta) = R_z^\top(\theta) = R_z(-\theta). \tag{7.4}$$

Thus an unexplained sign difference between two rotation matrices often comes from using active and passive conventions, respectively [34]. Figure 7 summarizes the two viewpoints.

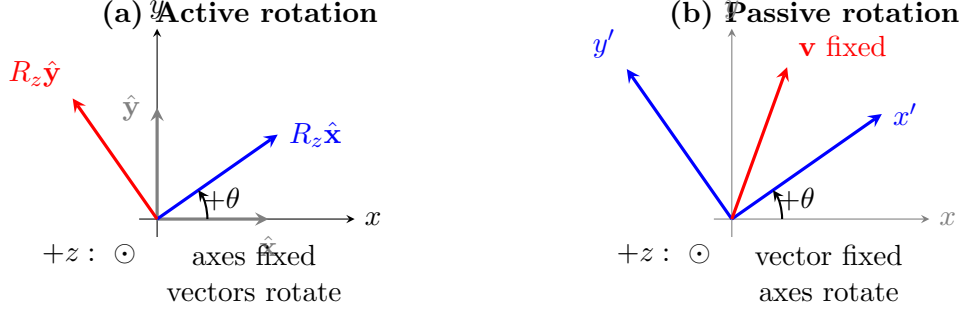


Figure 7: Original schematic comparing rotations about the z axis. (a) In an active rotation the coordinate axes remain fixed and the basis vectors rotate counterclockwise through $+\theta$; their images form the columns of $R_z(\theta)$. (b) In a passive rotation the physical vector remains fixed while the coordinate frame rotates through $+\theta$, so its coordinates transform with $R_z(-\theta) = R_z^\top(\theta)$. The symbol $\odot$ denotes the positive z direction out of the page. The convention follows the EE237 symmetry notes [34].

A spatial transformation that preserves lengths and angles is represented by an orthogonal matrix,

$$R^\top R = I. \quad (7.5)$$

Taking determinants of both sides gives

$$\det(R^\top R) = (\det R)^2 = 1, \quad \det R = \pm 1. \quad (7.6)$$

The sign distinguishes whether the transformation preserves or reverses the handedness of the coordinate system. In three dimensions, a proper rotation preserves handedness and has $\det R = +1$. A mirror reverses one direction and has $\det R = -1$. Full spatial inversion is represented by $R = -I_3$ and also has $\det R = (-1)^3 = -1$ [34].

For example, setting $\theta = 90^\circ$ gives

$$\hat{x} \rightarrow \hat{y}, \quad \hat{y} \rightarrow -\hat{x}, \quad \hat{z} \rightarrow \hat{z}, \quad (7.7)$$

and hence

$$R_z(90^\circ) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_z(90^\circ) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -y \\ x \\ z \end{pmatrix}. \quad (7.8)$$

Expanding the determinant along the first row,

$$\det R_z(90^\circ) = 0 - (-1) \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + 0 = +1. \quad (7.9)$$

Thus the matrix is a proper rotation [34].

For comparison, reflection across the xy plane and full spatial inversion are represented by

$$R_{xy} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad R_{\text{inv}} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (7.10)$$

Since the determinant of a diagonal matrix is the product of its diagonal entries,

$$\det R_{xy} = (1)(1)(-1) = -1, \quad \det R_{\text{inv}} = (-1)(-1)(-1) = -1. \quad (7.11)$$

The inversion result depends on dimension: $\det(-I_n) = (-1)^n$. In two dimensions, inversion therefore has determinant $+1$ and is equivalent to a 180° rotation; in three dimensions it has determinant -1 [34].

7.2 Polar and Axial Vectors

Position $\mathbf{r}$, momentum $\mathbf{p}$, electric field $\mathbf{E}$, and charge-current density $\mathbf{J}$ are polar vectors. Orbital angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, spin $\mathbf{S}$, magnetization $\mathbf{M}$, and magnetic field $\mathbf{B}$ are axial vectors. For an orthogonal spatial transformation represented by R , the two vector classes transform as

$$\mathbf{v}' = R\mathbf{v}, \quad \mathbf{a}' = (\det R)R\mathbf{a}, \quad (7.12)$$

where $\mathbf{v}$ is polar and $\mathbf{a}$ is axial. Thus they transform in the same way under proper rotations ($\det R = +1$) and differently under mirrors and inversion ($\det R = -1$) [34].

The most frequently used transformations are summarized below. The entries describe the vector attached to the spatial point after that point has been mapped by the operation.

Operation	$\mathbf{r}, \mathbf{p}$	$\mathbf{S}, \mathbf{M}$	$\mathbf{E}, \mathbf{B}$
Time reversal $\mathcal{T}$	$\mathbf{r} \rightarrow \mathbf{r}, \mathbf{p} \rightarrow -\mathbf{p}$	$\mathbf{S}, \mathbf{M} \rightarrow -\mathbf{S}, -\mathbf{M}$	$\mathbf{E} \rightarrow \mathbf{E}, \mathbf{B} \rightarrow -\mathbf{B}$
Inversion $\mathcal{P}$	$\mathbf{r}, \mathbf{p} \rightarrow -\mathbf{r}, -\mathbf{p}$	$\mathbf{S}, \mathbf{M} \rightarrow \mathbf{S}, \mathbf{M}$	$\mathbf{E} \rightarrow -\mathbf{E}, \mathbf{B} \rightarrow \mathbf{B}$
Mirror $\mathcal{M}_{\mathbf{n}}$	$\mathbf{v} \rightarrow \mathbf{v} - 2\mathbf{n}(\mathbf{n} \cdot \mathbf{v})$	$\mathbf{a} \rightarrow -\mathbf{a} + 2\mathbf{n}(\mathbf{n} \cdot \mathbf{a})$	polar $\mathbf{E}$; axial $\mathbf{B}$

Figure 8 visualizes the different sign rules for polar and axial vectors under spatial inversion.

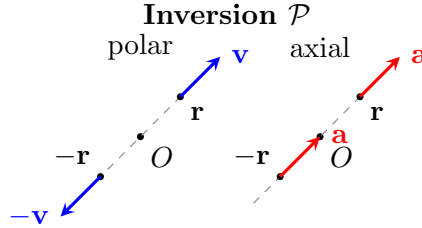


Figure 8: Original schematic of spatial inversion. Full inversion moves both vectors from $\mathbf{r}$ to $-\mathbf{r}$, reverses the polar vector, and preserves the axial-vector direction. The conventions follow the EE237 symmetry notes [34].

7.3 Time-Reversal Symmetry

Time reversal changes $t \rightarrow -t$ and reverses momenta and angular momenta. Spin is an intrinsic angular momentum, so

$$\mathcal{T}\mathbf{S}\mathcal{T}^{-1} = -\mathbf{S}, \quad \mathcal{T}i\mathcal{T}^{-1} = -i. \quad (7.13)$$

Here $\mathcal{T}$ is the time-reversal operator, not the superscript T used for a matrix transpose. Time reversal is antiunitary, so it complex-conjugates every scalar coefficient:

$$\mathcal{T}(c|\psi\rangle) = c^*\mathcal{T}|\psi\rangle. \quad (7.14)$$

Taking $c = i$, for which $i^* = -i$, and applying the operator product to an arbitrary state $|\phi\rangle$ gives

$$\mathcal{T}i\mathcal{T}^{-1}|\phi\rangle = -i\mathcal{T}\mathcal{T}^{-1}|\phi\rangle = -i|\phi\rangle. \quad (7.15)$$

Because this holds for every state, $\mathcal{T}i\mathcal{T}^{-1} = -i$. Thus the identity expresses complex conjugation by time reversal; it is not an ordinary matrix sign change. For example, in the position representation $\mathbf{p} = -i\hbar\nabla$, so complex conjugation reverses momentum:

$$\mathcal{T}\mathbf{p}\mathcal{T}^{-1} = i\hbar\nabla = -\mathbf{p}. \quad (7.16)$$

For a classical magnetization trajectory, the corresponding transformed trajectory is

$$\mathbf{m}_{\mathcal{T}}(t) = -\mathbf{m}(-t). \quad (7.17)$$

A magnetic state can therefore break time-reversal symmetry even when its nonmagnetic crystal structure does not. A fixed applied magnetic field, a fixed current, damping, or another nonequilibrium drive must not be silently left unchanged when testing time reversal; these time-odd or irreversible ingredients generally prevent the driven equation from being time-reversal-invariant.

Source status. Reference needed. Equations (7.13)– (7.17), including the antiunitary derivation and the general invariance statement above, still need an approved reference.

7.4 Parity or Spatial-Inversion Symmetry

Here parity means full spatial inversion,

$$\mathcal{P} : \quad \mathbf{r} \rightarrow -\mathbf{r}. \quad (7.18)$$

For inversion, $R_{\text{inv}} = -I_3$ and $\det R_{\text{inv}} = -1$. Substitution into the axial-vector rule in Eq. (7.12) gives

$$\mathbf{a}' = (\det R_{\text{inv}})R_{\text{inv}}\mathbf{a} = (-1)(-I_3)\mathbf{a} = \mathbf{a}. \quad (7.19)$$

Consequently, the shorthand $\mathbf{S}, \mathbf{M} \rightarrow \mathbf{S}, \mathbf{M}$ means that inversion does not reverse the components or direction of these axial vectors. It does move each vector from its original position to the inversion-related position:

$$(\mathbf{r}, \mathbf{S}(\mathbf{r}), \mathbf{M}(\mathbf{r})) \rightarrow (-\mathbf{r}, \mathbf{S}(\mathbf{r}), \mathbf{M}(\mathbf{r})). \quad (7.20)$$

Equivalently, when the transformed textures are written as functions of the coordinate $\mathbf{r}$,

$$\mathbf{S}_{\mathcal{P}}(\mathbf{r}) = \mathbf{S}(-\mathbf{r}), \quad \mathbf{M}_{\mathcal{P}}(\mathbf{r}) = \mathbf{M}(-\mathbf{r}). \quad (7.21)$$

This transformation law alone does not imply $\mathbf{S}(\mathbf{r}) = \mathbf{S}(-\mathbf{r})$ or $\mathbf{M}(\mathbf{r}) = \mathbf{M}(-\mathbf{r})$. Those equalities hold only when the complete magnetic texture is itself invariant under inversion. More generally, the structure has inversion symmetry only when the inverted sites, chemical species, interactions, and spin pattern reproduce the original structure. This is distinct from particle-exchange symmetry. In non-Hermitian coupled-mode models the symbol $\mathcal{P}$ may instead denote exchange of two subsystems; its meaning must therefore be stated explicitly.

Source status. Reference needed. Equations (7.18) and (7.19)–(7.21), including the axial-spin convention under inversion, still need an approved general reference.

7.5 Mirror Symmetry

A mirror with unit normal $\mathbf{n}$ reverses the perpendicular component of a polar vector. An axial vector behaves oppositely: its components parallel to the mirror plane reverse, while its perpendicular component is unchanged according to Eq. (7.12) [34]. For the xy mirror,

$$(v_x, v_y, v_z) \rightarrow (v_x, v_y, -v_z), \quad (S_x, S_y, S_z) \rightarrow (-S_x, -S_y, S_z). \quad (7.22)$$

The transformed lattice and every axial spin must both be compared with the original magnetic structure. A lattice may therefore possess a mirror plane that is broken by its magnetic order.

7.6 Spin-Rotation Symmetry

Spatial rotations move the lattice, whereas spin rotations act in spin space. For the isotropic exchange Hamiltonian used in ASTRO,

$$\mathcal{H}_{\text{ex}} = \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (7.23)$$

a common spin rotation R_s leaves every dot product unchanged:

$$(R_s \mathbf{S}_i) \cdot (R_s \mathbf{S}_j) = \mathbf{S}_i \cdot \mathbf{S}_j. \quad (7.24)$$

The exchange term consequently has global spin-rotation symmetry when the couplings J_{ij} do not depend on spin direction [11]. Magnetic anisotropy, Dzyaloshinskii–Moriya interaction, dipolar coupling, an applied field, or another spin–orbit-coupled term can reduce this continuous symmetry to the subset of joint spatial and spin operations that preserves the complete Hamiltonian.

7.7 Combined Parity–Time Symmetry in a Two-Sublattice Magnet

For the two-sublattice order parameters defined in Eq. (6.3), suppose that inversion exchanges sublattices A and B . Since inversion does not reverse an axial spin, whereas time reversal does, their actions are

$$\begin{array}{c|cc} & \mathbf{m} & \mathbf{n} \\ \hline \mathcal{P} & \mathbf{m} & -\mathbf{n} \\ \mathcal{T} & -\mathbf{m} & -\mathbf{n} \\ \mathcal{PT} & -\mathbf{m} & \mathbf{n} \end{array} \quad (7.25)$$

Thus a compensated antiferromagnetic pattern can be invariant under the combined operation even when it is not invariant under $\mathcal{P}$ or $\mathcal{T}$ separately. In a different use of parity, Zhang *et al.* define $\mathcal{P}$ as interchange of two coupled ferrimagnetic layers and obtain parity–time-symmetric Néel-vector dynamics at angular-momentum compensation [16]. These two meanings should not be conflated.

Source status. Reference needed. Equation (7.25) is a direct derivation from the provisional general transformation rules above. The cited paper supports only the separate coupled-layer example.

7.8 Mini-Tutorial: From $\mathcal{P}$, $\mathcal{T}$, and $\mathcal{PT}$ to Spintronic Effects

The useful question is not merely whether a crystal “has symmetry.” For a candidate response

$$O_i = \chi_{ij} F_j, \quad (7.26)$$

one must transform the observable O , the drive F , the magnetic order, and the complete device geometry. A symmetry of the driven structure is allowed to keep only tensor components that reproduce Eq. (7.26) after the transformation. Symmetry therefore decides which components may be nonzero, but not their magnitude or microscopic origin [21, 23, 24].

The signatures needed below follow directly from the polar/axial distinction:

Quantity	$\mathcal{P}$	$\mathcal{T}$	Character
$\mathbf{k}, \mathbf{E}, \mathbf{j}$	odd	$\mathbf{k}, \mathbf{j}$ odd; $\mathbf{E}$ even	polar
$\mathbf{S}, \boldsymbol{\sigma}, \mathbf{M}, \mathbf{B}$	even	odd	axial
spin current $Q_i^a \sim v_i S_a$	odd	even	polar flow index i , axial spin index a
Hall vector $\mathbf{h}$ in $\mathbf{j}_H = \mathbf{h} \times \mathbf{E}$	even	odd	axial response

The electric field is time-reversal even, whereas the dissipative current that it produces is time-reversal odd. A dc transport coefficient must therefore be interpreted as a nonequilibrium response, not as a new equilibrium order parameter.

Band-degeneracy checkpoint. Both $\mathcal{P}$ and $\mathcal{T}$ reverse $\mathbf{k}$, so their product leaves each crystal momentum fixed. For spin-1/2 electrons with $(\mathcal{PT})^2 = -1$, $\mathcal{PT}$ produces a Kramers-like partner at the same $\mathbf{k}$. Consequently, a $\mathcal{PT}$ -symmetric collinear magnet cannot have a same- $\mathbf{k}$ spin splitting. Breaking $\mathcal{PT}$ is necessary for such a splitting, but it is not sufficient: another unitary or antiunitary symmetry may still protect degeneracy [28, 30].

Source status. Reference needed. The response-tensor and $\mathcal{PT}$ band-degeneracy statements above are supported by the cited local reviews, but none of those reviews belongs to the project's approved reference classes. They are retained provisionally rather than replaced by a weaker indirect source.

7.8.1 Rashba Spin–Orbit Coupling: Break $\mathcal{P}$, Keep $\mathcal{T}$

For an interface or polar crystal with polar axis $\mathbf{n}$, the leading isotropic momentum-linear term is

$$\mathcal{H}_R = \alpha_R(\mathbf{n} \times \mathbf{k}) \cdot \boldsymbol{\sigma}. \quad (7.27)$$

Time reversal changes both $\mathbf{k}$ and $\boldsymbol{\sigma}$ in sign, so Eq. (7.27) is $\mathcal{T}$ even. Inversion changes $\mathbf{k}$ but not the axial spin; moreover an inversion-symmetric crystal cannot possess the fixed polar axis $\mathbf{n}$. Thus a nonzero Rashba coefficient requires inversion asymmetry, while time reversal may remain a symmetry. The eigenvalues

$$\varepsilon_{\pm}(\mathbf{k}) = \varepsilon_0(\mathbf{k}) \pm \alpha_R |\mathbf{n} \times \mathbf{k}| \quad (7.28)$$

are split at a general $\mathbf{k}$ but obey $\varepsilon_+(\mathbf{k}) = \varepsilon_-(-\mathbf{k})$, as required by $\mathcal{T}$ [20].

Source status. Reference needed. Equations (7.27) and (7.28) still need an approved project reference.

7.8.2 Dzyaloshinskii–Moriya Interaction: Break Bond-Center Inversion

For one bond,

$$\mathcal{H}_{ij}^{\text{DMI}} = -\mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j). \quad (7.29)$$

The directed bond and DMI-vector convention are shown in Fig. 1. If inversion about the bond midpoint exchanges i and j , the axial spins do not reverse but their ordered cross product does:

$$\mathcal{P}: \quad \mathbf{S}_i \times \mathbf{S}_j \longrightarrow \mathbf{S}_j \times \mathbf{S}_i = -\mathbf{S}_i \times \mathbf{S}_j. \quad (7.30)$$

Invariance would then require $\mathbf{D}_{ij} = -\mathbf{D}_{ij}$, hence $\mathbf{D}_{ij} = 0$. DMI therefore needs the relevant local bond-center inversion to be absent; the whole crystal need not be globally polar. Under time reversal both spins change sign, leaving their cross product and DMI unchanged. DMI is consequently compatible with $\mathcal{T}$ and is selected primarily by spatial symmetry [11, 3].

7.8.3 Spin Hall and Anomalous Hall Effects: Similar Geometry, Different $\mathcal{T}$ Requirement

In an isotropic spin Hall material the leading response can be written

$$Q_i^a = \sigma_{\text{SH}} \epsilon_{iaj} E_j. \quad (7.31)$$

Both Q_i^a and E_j are odd under inversion, while both are even under time reversal. Neither $\mathcal{P}$ nor $\mathcal{T}$ must therefore be broken for the spin Hall effect; crystal mirrors and rotations only restrict the allowed tensor components. This is why a centrosymmetric nonmagnetic heavy metal can generate a transverse spin current [5, 6, 7, 22].

For the anomalous Hall effect,

$$\mathbf{j}_{\text{AH}} = \mathbf{h} \times \mathbf{E}, \quad (7.32)$$

where the Hall vector $\mathbf{h}$ is axial and time-reversal odd. Inversion does not forbid it, but $\mathcal{T}$ does. Because $\mathcal{PT}$ also changes $\mathbf{h}$ in sign while leaving its spatial point fixed, $\mathcal{PT}$ forbids the anomalous Hall vector. Conversely, breaking $\mathcal{T}$ and $\mathcal{PT}$ is only a necessary screen: a mirror, rotation, or another antiunitary symmetry can still force a particular component of $\mathbf{h}$ to vanish. Compensated magnets can therefore have an anomalous Hall effect without a net moment when their magnetic point group allows a time-odd axial vector [27].

Source status. Reference needed. The anomalous-Hall symmetry criterion and Eq. (7.32) still need an approved project reference. Equation (7.31) is supported by the cited approved compact-model repository and the experimental references already used by this project.

7.8.4 Edelstein Effect: Uniform Versus Staggered Spin Accumulation

The Edelstein or inverse spin-galvanic response is

$$S_i = \chi_{ij} E_j. \quad (7.33)$$

Spin is inversion even and the electric field is inversion odd, so a global uniform tensor χ_{ij} must vanish when inversion is a symmetry. In a polar film with approximate $C_{\infty v}$ symmetry, the leading form is

$$\mathbf{S} = \chi_E \mathbf{n} \times \mathbf{E}. \quad (7.34)$$

The underlying material may preserve time reversal: the spin density is a nonequilibrium, current-odd response rather than an equilibrium magnetization.

Global inversion does not rule out a *staggered* Edelstein effect. If inversion exchanges two locally noncentrosymmetric magnetic sites, symmetry can impose

$$\mathbf{S}_A = -\mathbf{S}_B, \quad (7.35)$$

so that the unit-cell-average spin vanishes while a Néel-order-effective staggered field survives. This is the key distinction between global and local inversion asymmetry [23, 26, 21].

Source status. Reference needed. The uniform and staggered Edelstein equations need an approved project reference.

7.8.5 Spin–Orbit Torque: Convert a Symmetry-Allowed Spin Response into Torque

Once a current produces a spin polarization $\boldsymbol{\zeta}$, exchange coupling to a magnetic order permits the two familiar lowest-order torques

$$\boldsymbol{\tau}_{\text{SOT}} = \tau_{\text{FL}} \mathbf{m} \times \boldsymbol{\zeta} + \tau_{\text{DL}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\zeta}). \quad (7.36)$$

The shared device axes and the two torque vector factors are shown in Fig. 3. The source symmetry determines $\boldsymbol{\zeta}$: a conventional isotropic bilayer gives $\boldsymbol{\zeta} \parallel \mathbf{n} \times \mathbf{E}$, whereas lower crystal symmetry allows additional components. Field-like and damping-like angular forms do not uniquely identify Rashba/Edelstein and spin Hall microscopic mechanisms, because either mechanism can contribute to both forms [21, 4, 9].

A single vertical mirror illustrates how an unconventional component appears. Take the surviving mirror to be the yz plane. Then $E_x \rightarrow -E_x$, while the axial component $\zeta_z \rightarrow -\zeta_z$, so $\zeta_z \propto E_x$ is allowed. For current inside the mirror plane, $E_y \rightarrow E_y$ but $\zeta_z \rightarrow -\zeta_z$, so the same response is forbidden. This is the symmetry logic behind the current-direction-dependent out-of-plane torque in WTe₂/ferromagnet bilayers [34, 25, 24].

For antiferromagnets, global inversion can forbid the uniform torque while local inversion asymmetry permits opposite sublattice torques. Whether the even or odd response is staggered depends on the operation that exchanges the sublattices; local noncentrosymmetry is necessary but not sufficient, so the full site and magnetic point groups must be checked [23, 26].

7.8.6 Spin-Transfer Torque: No Intrinsic Inversion-Breaking Requirement

Spin-transfer torque transfers angular momentum from a spin-polarized current whose polarization $\mathbf{p}$ is prepared by a magnetic polarizer, magnetic texture, or spin filter:

$$\boldsymbol{\tau}_{\text{STT}} = u_{\text{DL}} \mathbf{m} \times (\mathbf{p} \times \mathbf{m}) + u_{\text{FL}} \mathbf{m} \times \mathbf{p}. \quad (7.37)$$

Unlike Rashba SOC, DMI, or a uniform Edelstein response, conventional STT does not require spin-orbit coupling or broken crystal inversion. Its essential ingredients are a nonequilibrium spin flux and a noncollinear receiving magnetization. Magnetic order or spin filtering supplies the time-reversal-odd polarization $\mathbf{p}$, while reversing the current reverses the angular-momentum flux. Device geometry can of course break inversion, but that is not the microscopic selection rule for STT [5, 13, 12].

7.8.7 Altermagnetic Spin Splitting: Break $\mathcal{T}$ Without $\mathcal{PT}$ Degeneracy

In a conventional collinear antiferromagnet, opposite-spin sublattices are often related by inversion or a translation. A $\mathcal{PT}$ symmetry then protects same- $\mathbf{k}$ degeneracy directly. In the nonrelativistic spin group, a spin flip combined with a sublattice translation can likewise make the two spin channels degenerate throughout the Brillouin zone; after spin-orbit coupling is included, translation-time-reversal alone generally protects degeneracy only at selected momenta. An altermagnet instead relates the opposite-spin sublattices by a proper or improper crystal rotation $\mathcal{R}$, but not by a translation or inversion based spin-group operation that would restore general same- $\mathbf{k}$ degeneracy. Schematically,

$$\varepsilon_{\uparrow}(\mathbf{k}) = \varepsilon_{\downarrow}(\mathcal{R}\mathbf{k}), \quad \varepsilon_{\uparrow}(\mathbf{k}) \neq \varepsilon_{\downarrow}(\mathbf{k}) \quad \text{at a general } \mathbf{k}. \quad (7.38)$$

The rotation redistributes the two spin channels so that the net moment can still compensate. The resulting even-parity d -, g -, or i -wave spin splitting is nonrelativistic and does not require spin-orbit coupling; spin-orbit coupling subsequently modifies nodal degeneracies and enables additional Berry-curvature responses. Inversion may still exist as an operation within a same-spin sector, but it must not exchange the two opposite-spin sublattices in the way that restores $\mathcal{PT}$ degeneracy [28, 29, 30].

Source status. Reference needed. Equation (7.38) and the altermagnetic symmetry classification still need an approved project reference.

7.8.8 Selection-Rule Summary

Effect	Spatial-symmetry gate	Time-reversal / combined-symmetry gate
Rashba SOC	Global inversion broken; a polar axis is allowed	$\mathcal{T}$ may be preserved; $\mathcal{PT}$ degeneracy is absent
DMI	Inversion absent at the relevant bond midpoint	$\mathcal{T}$ may be preserved
Spin Hall effect	$\mathcal{P}$ may be preserved; point group selects tensor components	$\mathcal{T}$ may be preserved
Anomalous Hall effect	$\mathcal{P}$ may be preserved; remaining mirrors and rotations select $\mathbf{h}$	$\mathcal{T}$ broken and no $\mathcal{PT}$ or other symmetry that reverses the allowed Hall vector
Edelstein effect	Uniform response needs global inversion breaking; staggered response needs local inversion breaking	Equilibrium $\mathcal{T}$ breaking is not required for the driven spin accumulation
SOT	Spin-source and device point group must allow ζ ; local asymmetry can give staggered AFM SOT	Magnetic target breaks $\mathcal{T}$; torque parity alone does not identify the source mechanism
STT	No intrinsic $\mathcal{P}$ -breaking or SOC requirement	Requires a nonequilibrium spin flux; magnetic polarizer/filter supplies time-odd $\mathbf{p}$
Altermagnetic splitting	Opposite-spin sublattices rotation-related, not inversion- or translation-related	$\mathcal{T}$ broken; no $\mathcal{PT}$ or translation/inversion based spin-group symmetry that enforces general same- $\mathbf{k}$ degeneracy

This table gives necessary symmetry gates, not sufficient material-design conditions. After passing the gate, band filling, spin-orbit strength, scattering, interfaces, magnetic domains, and device geometry determine whether the effect is measurable or useful.

7.9 Practical Symmetry Test

To test a proposed symmetry of a spin model:

1. map every position and sublattice label;
2. transform spin as an axial vector and reverse it additionally under time reversal;
3. transform external fields, currents, and boundary conditions;
4. rebuild every Hamiltonian or dynamical term with the transformed quantities; and
5. compare the complete transformed model and magnetic pattern with the originals.

Checking only the atomic lattice or only the spin arrows is insufficient for a magnetic symmetry.

Source status. Reference needed. The general five-step test is a provisional pedagogical synthesis and still needs an approved reference.

References

- [1] J. Park, G. E. Rowlands, O. J. Lee, D. C. Ralph, R. A. Buhrman, Macrospin modeling of sub-ns pulse switching of perpendicularly magnetized free layer via spin-orbit torques for cryogenic memory applications, Appl. Phys. Lett., **105**, 102404 (2014).

- [2] R. F. L. Evans, W. J. Fan, P. Chureemart, T. A. Ostler, M. O. A. Ellis, R. W. Chantrell, Atomistic spin model simulations of magnetic nanomaterials, *J. Phys.: Condens. Matter*, **26**, 103202 (2014).
- [3] Z. Kong, Z. Xu, X. Zhang, B. Zhao, Y. Yang, L. Zhu, Z. Zhu, Spatially asymmetric switching under spin-orbit torque and Dzyaloshinskii-Moriya interaction, *J. Appl. Phys.*, **140**, 063903 (2026).
- [4] X. Zhang, Z. Xu, Z. Zhu, Revisiting the analytical solution of spin-orbit torque switched nanoscale perpendicular ferromagnets, *Phys. Rev. B*, **110**, 184428 (2024).
- [5] Z. Zhu, SOT_STT_compact_model compact SOT/STT device model, GitHub repository, commit 1df8947.
- [6] L. Liu, C.-F. Pai, Y. Li, H. W. Tseng, D. C. Ralph, R. A. Buhrman, Spin-torque switching with the giant spin Hall effect of tantalum, *Science*, **336**, 555–558 (2012).
- [7] C.-F. Pai, L. Liu, Y. Li, H. W. Tseng, D. C. Ralph, R. A. Buhrman, Spin transfer torque devices utilizing the giant spin Hall effect of tungsten, *Appl. Phys. Lett.*, **101**, 122404 (2012).
- [8] Z. Xu, X. Zhang, Y. Qiao, G. Liang, S. Shi, Z. Zhu, Deterministic spin-orbit torque switching including the interplay between spin polarization and kagome plane in Mn_3Sn , *Phys. Rev. B*, **109**, 134433 (2024).
- [9] Z. Xu, J. Ren, Z. Yuan, Y. Xin, X. Zhang, S. Shi, Y. Yang, Z. Zhu, Field-free spin-orbit torque switching of an antiferromagnet with perpendicular Néel vector, *J. Appl. Phys.*, **133**, 153904 (2023).
- [10] Z. Xu, Y. Zhou, X. Zhang, Y. Qiao, Z. Xu, D. Shao, Z. Zhu, Spin-orbit torque-driven chiral domain wall motion in Mn_3Sn , *Appl. Phys. Lett.*, **127**, 072402 (2025).
- [11] Z. Zhu, ASTRO atomistic spin-dynamics code, GitHub repository, commit 9c807cf.
- [12] Z. Zhu, J. Deng, X. Fong, G. Liang, Voltage-input spintronic oscillator based on competing effect for extended oscillation regions, *J. Appl. Phys.*, **125**, 183902 (2019).
- [13] Z. Yuan, J. Long, Z. Xu, Y. Xin, L. An, J. Ren, X. Zhang, Y. Yang, Z. Zhu, Anomalous impact of thermal fluctuations on spin transfer torque induced ferrimagnetic switching, *J. Appl. Phys.*, **133**, 153903 (2023).
- [14] X. Zhang, B. Cai, J. Ren, Z. Yuan, Z. Xu, Y. Yang, G. Liang, Z. Zhu, Spatially nonuniform oscillations in ferrimagnets based on an atomistic model, *Phys. Rev. B*, **106**, 184419 (2022).
- [15] Z. Zhu, K. Cai, J. Deng, V. P. K. Miriyala, H. Yang, X. Fong, G. Liang, Electrical generation and detection of terahertz signal based on spin-wave emission from ferrimagnets, *Phys. Rev. Applied*, **13**, 034040 (2020).
- [16] X. Zhang, C. Liu, Z. Zhu, Y. Zhang, Non-Hermitian spin dynamics in a coupled ferrimagnetic system, *Phys. Rev. B*, **109**, 184426 (2024).
- [17] H. Guo, Z. Lin, J. Lu, C. Yun, G. Han, S. Sun, Y. Wu, W. Yang, D. Xiao, Z. Zhu, L. Peng, Y. Ye, Y. Hou, J. Yang, Z. Luo, Layer-dependent spin-orbit torque switching of Néel vector in a van der Waals antiferromagnet, *Nat. Commun.*, **16**, 8911 (2025).

- [18] Z. Wang, Y. Xiang, R. Chen, Z. Sun, C. Hong, X. Chen, J. Gao, S. Wu, Z. Guo, Y. Chen, Q. Mi, Z. Liu, S. Yan, H. Lei, W. Ruan, Y. Zhang, W. Yu, W.-T. Liu, Z. Yuan, S. Wu, Ferromagnet-like binary switching of a Stoner–Wohlfarth antiferromagnet, *Nature*, **650**, 340–345 (2026).
- [19] F. Xue, P. M. Haney, Intrinsic staggered spin-orbit torque for the electrical control of antiferromagnets: Application to CrI_3 , *Phys. Rev. B*, **104**, 224414 (2021).
- [20] A. Manchon, H. C. Koo, J. Nitta, S. M. Frolov, R. A. Duine, New perspectives for Rashba spin–orbit coupling, *Nat. Mater.*, **14**, 871–882 (2015).
- [21] A. Manchon, J. Železný, I. M. Miron, T. Jungwirth, J. Sinova, A. Thiaville, K. Garello, P. Gambardella, Current-induced spin–orbit torques in ferromagnetic and antiferromagnetic systems, *Rev. Mod. Phys.*, **91**, 035004 (2019).
- [22] A. Davidson, V. P. Amin, W. S. Aljuaid, P. M. Haney, X. Fan, Perspectives of electrically generated spin currents in ferromagnetic materials, *Phys. Lett. A*, **384**, 126228 (2020).
- [23] J. Železný, H. Gao, A. Manchon, F. Freimuth, Y. Mokrousov, J. Zemen, J. Mašek, J. Sinova, T. Jungwirth, Spin–orbit torques in locally and globally noncentrosymmetric crystals: Antiferromagnets and ferromagnets, *Phys. Rev. B*, **95**, 014403 (2017).
- [24] Y. Liu, Q. Shao, Two-dimensional materials for energy-efficient spin–orbit torque devices, *ACS Nano*, **14**, 9389–9407 (2020).
- [25] D. MacNeill, G. M. Stiehl, M. H. D. Guimarães, R. A. Buhrman, J. Park, D. C. Ralph, Control of spin–orbit torques through crystal symmetry in WTe_2 /ferromagnet bilayers, *Nat. Phys.*, **13**, 300–305 (2017).
- [26] H. Watanabe, Y. Yanase, Symmetry analysis of current-induced switching of antiferromagnets, *Phys. Rev. B*, **98**, 220412(R) (2018).
- [27] L. Šmejkal, A. H. MacDonald, J. Sinova, S. Nakatsuji, T. Jungwirth, Anomalous Hall antiferromagnets, *Nat. Rev. Mater.*, **7**, 482–496 (2022).
- [28] L. Šmejkal, J. Sinova, T. Jungwirth, Beyond conventional ferromagnetism and antiferromagnetism: A phase with nonrelativistic spin and crystal rotation symmetry, *Phys. Rev. X*, **12**, 031042 (2022).
- [29] L. Šmejkal, J. Sinova, T. Jungwirth, Emerging research landscape of altermagnetism, *Phys. Rev. X*, **12**, 040501 (2022).
- [30] T. Jungwirth, J. Sinova, R. M. Fernandes, Q. Liu, H. Watanabe, S. Murakami, S. Nakatsuji, L. Šmejkal, Symmetry, microscopy and spectroscopy signatures of altermagnetism, *Nature*, **649**, 837–847 (2026).
- [31] Z. Zhu, ODE_LLG integral macrospin LLGS code, GitHub repository, commit **348e46c**.
- [32] EE237 Week 5 lecture slides, Magnetic Anisotropy, Magnetostatics, and Zeeman Energy, slides 13–14 and 21–22.
- [33] EE237 Week 10 lecture slides, MRAM Cells, Reliability, and Design Tradeoffs, slide 28.
- [34] EE237 Week 11 lecture slides, Spin–Orbit Torques and Switching Symmetry, slides 4–5.